Hugo Armando Guillén-Ramírez

Curriculum Vitae

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Objective

I am Ph.D. candidate working with machine learning applied to RNA bioinformatics. My research interests are mostly machine learning, algorithm design, software development, optimization using bio-inspired strategies, bioinformatics, and graph theory applied to bioinformatics.

Education

2014–Current **PhD in Computer Science**, *CICESE Research Center*, Ensenada, Baja California, Mexico, Grade: 97.02/100.

2011–2014 **MSc in Computer Science**, *CICESE Research Center*, Ensenada, Baja California, Mexico, Grade: 96.78/100.

2007–2011 **Bachelor in Computer Systems Engineering**, *Instituto Tecnológico de Tepic*, Nayarit, Mexico, Grade: 97/100.

Doctoral thesis (ongoing)

Title *Methods for non-coding RNA gene prediction* (translated from Spanish)

Description The main objective of this project is to develop a non-coding RNA (ncRNA) gene predictor based in homology search on known sequences supported by machine learning, meta-heuristics, and state-of-the-art biological information.

Expected graduation date: December 2019

Master thesis

Title *Design of a Tissue P-System and a molecular algorithm to solve the MAX-CLIQUE problem* (translated from Spanish)

Description This theoretical work presents a tissue P-System that solves the MAX-CLIQUE problem in polynomial time, and also a constructive DNA algorithm inspired by this system.

Languages

Spanish Native

English Advanced

Skills

Advanced	PYTHON, Weka 3
Intermediate	SQL (PostgreSQL, MySQL, SQLite), JAVA, C#, PHP, XHTML, CSS, JAVASCRIPT, LaTeX, LINUX
Basic	C++, VISUALBASIC .NET

Publications

Journal	**Hugo A. Guillén-Ramírez**, Israel M. Martínez-Pérez. "Classification of riboswitch sequences using k-mer frequencies". In Biosystems. Elsevier, 2018. [Accepted, In Press]
Conference	Roilhi F. Ibarra-Hernández, Nancy Bertin, Miguel A. Alonso-Arévalo, **Hugo A. Guillén-Ramírez**. "A benchmark of heart sound classification systems based on sparse decompositions". In 14th International Symposium on Medical Information Processing and Analysis (SIPAIM 2018), Mexico. [Accepted]
Conference	**Hugo A. Guillen-Ramirez**, Jose Colbes, Carlos A. Brizuela, Israel M. Martinez-Perez. "Accurate Classification of Immunomodulatory RNA Sequences". In International Joint Conference on Neural Networks (IJCNN) 2017. pp. 236-241. IEEE, 2017.
Poster	**Hugo A. Guillen** and Israel Martinez-Perez. "Simulation of a programmable molecular computer". Presented in the 21st International Conference on DNA Computing and Molecular Programming 2015, Boston, USA.
Conference	**Hugo A. Guillen** and Israel Martinez-Perez. "Challenges in the prediction of non-coding RNA genes" (translated from Spanish). In the proceedings of the National Meeting on Computer Science (Encuentro Nacional de Ciencias de la Computación, ENC 2014), Mexico.
Conference	Brizuela, Carlos A., Jorge E. Luna-Taylor, Israel Martinez-Perez, **Hugo A. Guillen**, David O. Rodriguez, Armando Beltran-Verdugo. "Improving an evolutionary multi-objective algorithm for the biclustering of gene expression data". In Evolutionary Computation (CEC), 2013 IEEE Congress on, pp. 221-228. IEEE, 2013.
Poster	**Hugo A. Guillen** and Israel Martinez-Perez. "Design of a Tissue P-System for solving the MAX-CLIQUE problem" (translated from Spanish). Presented in the 6th International Congress on Computer Science (6to.Congreso Internacional en Ciencias Computacionales, CiComp 2013), Mexico.

Experience

Developer

2013–Current	During my doctoral work I've been programming workflows in Python 3, integrating and maintaining research software for machine learning and data visualization in Python, MySQL, SQLite, and Weka for machine learning tasks.
2017	Developer of NB2PY, a Python 3 open source package that exports cells from Jupyter Notebooks to plain text files: `https://pypi.org/project/nb2py/`.

2016 Developer of REFRACTIVEINDEX.INFO-SQLITE, a Python 3 open source package that retrieves information from the refractiveindex.info database locally. `http://github.com/HugoGuillen/refractiveindex.info-sqlite`

2012 MXGlobal Solutions - Tepic, Nayarit, Mexico. I developed C# and SQL software.

2010-2011 Instituto Tecnológico de Tepic - Tepic, Nayarit, Mexico. I developed a web system in Java/JSP.

Teaching

2017–Current Teacher of the **Artificial Intelligence**, **Advanced Mathematics**, **Evolutive Computing**, and **Assembly Language** courses at Universidad Autónoma de Baja California, Ensenada, Mexico.

2017–2018 Taught **Mathematics II, IV, and V** (in English), **Intermediate Algebra I and II** (in English) and **Mathematical reasoning V and VI** (in Spanish) at Xochicalco High School, Ensenada, Mexico.

2012 T.A. of the **Discrete Mathematics** course at CICESE Research Center.

Talks

2017, Mar Seminar **Score functions for discriminant k-mers and their application in ncRNA classification** (translated from Spanish) at CICESE Research Center, Mexico.

2016, Oct Seminar **Methods for ncRNA gene prediction: RNA secondary structure alternative graph representations** (translated from Spanish) in the XII National Meeting on Mathematical Biology (Center of Mathematical Sciences, UNAM Campus Morelia), Mexico.

2012, May Seminar **Introduction to genetic algorithms** (translated from Spanish) at Instituto Tecnológico de Tepic, Mexico.

Attendant

2016 XVIII Fall School on Mathematical Biology (Center of Mathematical Sciences, UNAM Campus Morelia), Mexico.

2016 NGS Data Analysis and Informatics Conference, San Diego, USA.

2015 Applications of Artificial Organic Networks (National Meeting on Computer Science 2015, Mexico).

Scholarships

2014 CONACYT National Scholarship for doctoral studies (Mexico).

2011 CONACYT National Scholarship for master studies (Mexico).

References

Available at request.